

Evolution of Firewalls: Toward Securer Network Using Next Generation Firewall

Junyan Liang

Department of Computer Science
University of Nevada Las Vegas
Las Vegas, NV 89154, USA
liangj10@unlv.nevada.edu

Yoohwan Kim

Department of Computer Science
University of Nevada Las Vegas
Las Vegas, NV 89154, USA
Yoohwan.Kim@unlv.edu

Abstract—The Internet was not created with security provisions and, as a result, many security protocols and devices were added later. Firewall is an essential device that gives the basic protection to the networks and devices connected to the Internet. The first generation firewalls were adopted in the early 1980s and operated based on a set of simple rules that controlled the network traffic between the Internet and the local network. Then over the years, firewalls have evolved into many different kinds to provide better protections. However, traditional firewalls are not catching up with the evolving threats, and the industry is rapidly adopting the Next-Generation Firewall (NGFW) that offer higher performance, better protection, and simplicity. This paper will review the evolution of the traditional firewalls, the need for NGFW, its features, and the industrial adoption trends. We will also review how NGFW can support emerging machine learning algorithms and better protect IoT systems.

Keywords—NGFW, IPS, Firewall, Deep Packet Inspection, Cyber Security, Network Threats

I. INTRODUCTION

The very first firewall was designed and developed by the Digital Equipment Cooperation (DEC) back in the 1980s [1]. These firewalls operate at the network layer (layer 3) of the Open Systems Interconnection (OSI) model [1]. It operates based on a set of rules, in which packets are filtered by their source address, destination address, and port numbers. However, with the rapid evolution of applications and threats, the traditional firewalls are becoming less and less capable of adequately protecting both personal and corporate networks [2]. There were over 7.9 billion records exposed just between January and September in 2019, which is a 33% increase from the same time in 2018 [3]. Thus, newer types of firewalls were invented. Each one of them was designed to deal with specific threats. There are generally five types of firewalls, packet filtering firewall, application-gateway firewall, circuit-gateway firewall, stateful-firewall, and next-generation firewall [4]. The history, the functionalities, the vulnerabilities, and the features of the first four types of firewalls will be discussed briefly in section II. In addition, web application firewall (WAF) and unified threat management (UTM) will also be mentioned in section II. In section III, we will go over the history and the importance of NGFW. Some of the critical features of NGFW will be examined in this section as well, such as deep packet inspection and intrusion prevention system (IPS). In section IV, some of the recent applications and research in the field of NGFW will be discussed. Finally, section V will be the summary of the survey.

II. TYPES OF FIREWALLS

A. Firewall Evolution

The very first generation firewall was the packet filtering firewall. It was developed by Digital Equipment Cooperation (DEC) in the year of 1988. It was not aimed to detect a virus or any network threats, it was only used to protect the network against unwanted packets from certain source IP addresses. In 1990, AT&T Bell Labs developed the stateful firewall which aims to make packet filtering networks more efficient. As the internet slowly getting more and more popular, the application-gateway firewall was developed in 1991 by DEC, which was meant to detect network attacks. Then in 1997, the web application firewall was introduced to protect web servers, and it was a huge success when it comes to detecting network attacks. After almost a decade later, the concept of UTM was introduced in 2004. The concept of UTM was to combine every type of firewall into one big machine. In 2009, the concept of UTM evolved into the concept NGFW. In later sections, the difference between UTM and NGFW will be discussed in detail. After a whole decade later, NGFW becomes the most popular choice when it comes to defending industrial and enterprises network, especially when it is combined with machine learning and other new and advanced intrusion detection techniques. In the next sections, each type of firewall will be explained and discussed in detail. Noted that all firewalls discussed in this paper are network firewalls, host firewalls will not be discussed here.

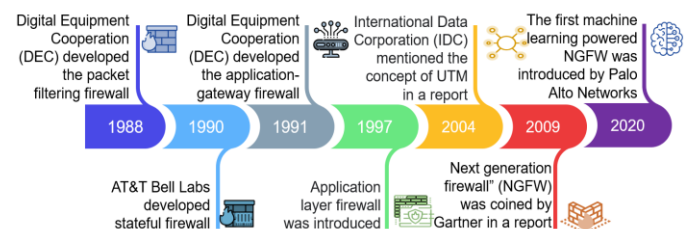


Fig. 1. Firewall evolution map

B. Traditional Firewalls

1) Packet Filtering Firewall

The packet filtering firewall is the very first firewall. It filters packets by monitoring the incoming and outgoing packets. It filters them by a set of rules, which are determined by the packets' Internet Protocol (IP) addresses, protocols, and ports.

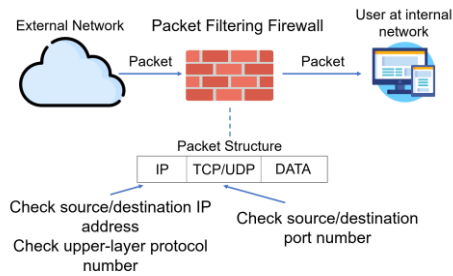


Fig. 2. Packet Filtering Firewall

Because of its basic functions, the packet filtering firewall is easy to configure and can filter packets efficiently. However, the packet filtering firewall does not check payload, which is located at the application layer of the OSI model. It also does not authenticate users and can be vulnerable to IP spoofing. Thus making this type of firewall not very secure against higher-level threats [5].

2) Stateful-Firewall

Stateful-firewall is the upgrade version of the packet filtering firewall. It operates based on the concept of the three-way handshake process in TCP. Stateful means that the firewall can keep track of established flows and all the packets that are inside the traffic flow in both directions. In order to achieve this, there is a cache inside the firewall, which allows it to keep records of every traffic flow. Whenever a new connection passes through, a syn-packet in TCP flow, a profile is created for this connection based on the header information of this traffic such as IP addresses and port numbers. Any subsequent connection that attempts later is checked against this list of records by the stateful firewall. If the connection does exist in the cache, then the packet is allowed to pass through the firewall. If the connection is not listed in the cache, then the stateful firewall will decide if this connection is qualified to be listed in the cache by using the basic packet filtering rules. The stateful firewall removes the entry in the cache if the connection is torn down, the FIN-packet of TCP arrived. Alternatively, if the connection is inactive for too long, and the entry of that connection is removed from the cache. Either way, the traffic flow is blocked because it is no longer listed in the cache [5].

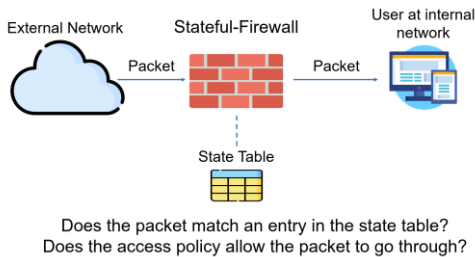


Fig. 3. Stateful-Firewall

The stateful firewall is more secure and more efficient than the packet filtering firewall. But it also requires more space due to its cache. However, it still has its unique disadvantages. For example, the cache table can be overflowed if there are too many traffic flows caused by a Denial of Service (DoS) attack. It still inherits the same flaws from the packet filtering firewall,

in which it does not check payload and does not authenticate users.

C. Application-Gateway Firewall

Unlike the previous firewalls mentioned. Application-gateway firewall provides application-level control over network traffic. Application-gateway firewall is no longer a “wall” that is in-between the source and the destination of the traffic. It is more like a “gateway,” a gateway that examines incoming packets at the application level and uses proxies to create sessions with remote users. A secure gateway is created whenever a client tries to access a server’s network resources. All incoming network resources from the server go to the gateway first then get sent back to the client if requested.

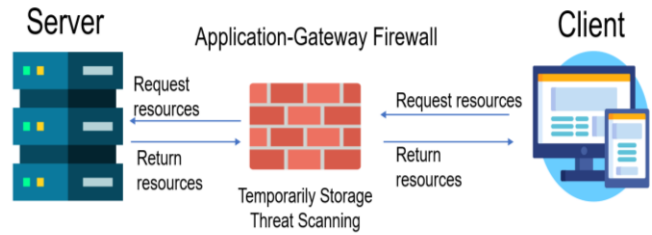


Fig. 4. Application-Gateway Firewall [6]

This provides an extra layer of protection, which prevents the client from directly accessing the server’s network resources. If there are any corrupted network resources, the gateway will be affected first, and be dealt with, hence not affecting the client [6].

However, this extra layer of protection does come at some cost. Namely, the speed of the traffic between user and server is generally slower due to the secure gateway in between them.

D. Circuit-Gateway Firewall

Circuit-gateway firewall is similar to an application-gateway firewall in the sense that it is like a “gateway” in-between the client and the server. Unlike application-gateway firewall, circuit-gateway firewall does not operate at the layer 7 (application layer) of the OSI model. Instead, it operates at the layer 5 (session layer) of the OSI model. But what does this all mean? Instead of focusing on the server’s network resources, the circuit-gateway firewall focuses on the connections between the client and server. A Circuit-gateway firewall provides a proxy that creates a transparent connection called a virtual circuit that acts as an end-to-end connection between the client and the server, and it relays TCP connections between the client and the server. By doing this, the circuit-gateway firewall ensures the connections are legitimate. It also makes sure that the responses match up with requests, i.e., what is coming out of the server matches up with the request from the client. If the responses do not match with the requests, then there is a chance someone is trying to exfiltrate data out of the network [4].

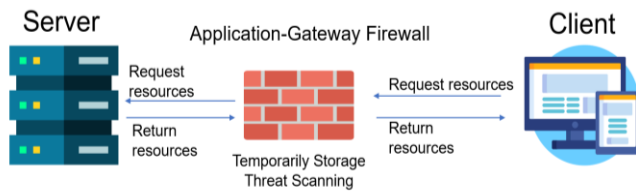


Fig. 5. Circuit-Gateway Firewall [6]

Because circuit-gateway firewall only operates at the session layer of the OSI model, it does not provide any packet-level defense. It only confirms the connections are legitimate, does not care what the payload is. Hence it is faster than an application-gateway firewall but less secure. It is often used with other firewalls.

E. Web-Application Firewall

Web-application firewall (WAF) is a particular type of application firewall that applies specifically to web applications. It is a security element in an application proxy device that protects the back-end web application server from different threats. It protects the webserver by inspecting the HTTP/HTTPS request packets and the network traffic patterns. If any malicious packets or patterns are found, WAF prevents the attacks by blocking the HTTP requests or the by destroying the session between client and server [7].

WAF is exceptionally effective against known web attacks, such as cross-site scripting (XSS), SQL injection, and DDoS attacks [8]. On the other hand, it is not capable of dealing with unknown vulnerabilities, aka zero-day exploits. Because the majority part of the threat detection mechanism from a WAF is based on pattern recognition. A never-seen attack is difficult for WAF to recognize because of the lack of pattern. Hence, it is difficult to defend against them. But zero-day exploits are not easy to find, and many websites suffer from known vulnerability attacks every day. That is why the market of WAF is growing steadily. According to IndustryARC, the WAF market was expected to grow to \$8.05 billion by 2025.[9]

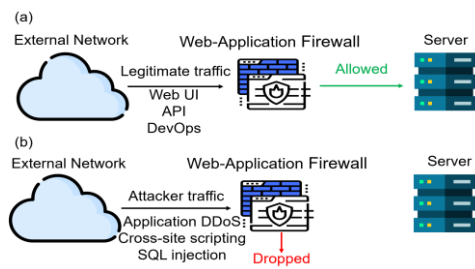


Fig. 6. Web-Application Firewall. (a) legitimate traffic (b) attacker traffic

F. Unified Threat Management

Unified Threat Management (UTM) is an approach in which a single software or hardware device provides multiple security functions that is capable of dealing with different types of cybernetic threats, hence the name unified threat management. UTM provides all sorts of defense mechanisms from the previously mentioned firewalls such as rule matching packet filtering, stateful inspection, deep packet inspection,

intrusion detection (or prevention) system, application, circuit gateway, and others ...

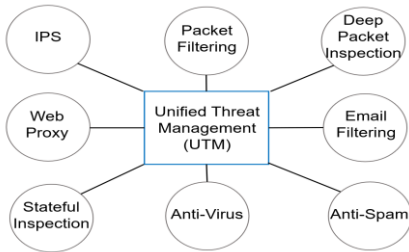


Fig. 7. Unified Threat Management [10].

In addition to the traditional firewall defense mechanisms, UTM also provides defense against spamming and phishing by complete content inspection. UTM reduces the complexity of maintaining multiple firewalls for different purposes, and it is easy to deploy and install. It is one of the first choices for industrial network protection [11].

However, due to its multi-layers of protection in a single device. It suffers from a single point of failure, where one part of the defense layers failed would affect the whole network. It also drastically decreases network performance because the packets have to go through multiple levels of inspection and examination [12].

G. Summary

In the previous sections, six types of firewalls were discussed. Each of them has distinct features, strengths and weaknesses. Table I describes which OSI layer these firewalls operate at, their advantages and their disadvantages.

TABLE I. FIREWALL COMPARISON

Types of Firewall	Firewall Features		
	OSI Model Layer	Advantages	Disadvantages
Packet Filtering Firewall	Network Layer (Layer 3)	Efficient Easy to set up	Do not inspect payload No user authentication
Stateful Firewall	Network Layer (Layer 3)	More efficient	Takes more space Vulnerable to DoS attack
Circuit-Gateway Firewall	Session Layer (Layer 5)	Defense in layer 5 Detects if data is being exfiltrated	Additional time to process traffic
Application-Gateway Firewall	Application Layer (Layer 7)	Defense in layer 7 Detect malicious payload	Additional time to process traffic
Web-Application Firewall (WAF)	Application Layer (Layer 7)	Defense against known web attacks	Unable to detect zero-day attacks
Unified Threat Management (UTM)	All Layers	All layer protection Additional protection, such as anti-spam	Single point of failure Could decrease performance drastically

III. NEXT-GENERATION FIREWALL

A. Threat Evolution

In the early years of the internet, the earliest network ‘hacks’ were not malicious in nature as they appear today. They were more like an experiment to test the limits of computer networks. As they grew more sophisticated, they became more recreational nature, more like a prank. These types of software can be bothersome but mostly not harmful. In fact, many infamous hackers before the 90s were hacking just to show that they could. As the influence of the internet grew, there were more opportunities for the opportunists to gain wealth and fame through the use of malicious software, or malware [13].

Today, there are many different types of cybernetic threats that we face daily. These threats manifest themselves in different ways. They can target directly to the operating system (OS), for example, malware, worms, viruses, trojan horses, and ransomware. Or they can target the communication channels between devices, for example, man-in-the-middle attacks, DDoS attacks. Or they can be the newer generation of internet threats like zero-day attacks and advanced persistent threats (APT).

With massive internet security breaches like Sony Pictures hack in 2014, the WannaCry ransomware cyber-attack in 2017, and the Zoom Credentials Hack in 2020. Cybercrimes are no longer news today. Cybercrime is so advanced, and the organizations behind these crimes are so sophisticated that it is impossible to prevent all cyber-attacks. All we can do are minimize the damages of these attacks brought by prevention and recovery with even more sophisticated protocols, training, and technologies. This is where NGFW can help.

B. What is NGFW

Next-Generation Firewall (NGFW) was invented around 2009 by Gartner. It was similar to the UTM in the early days and since then has evolved into a more innovative and more sophisticated product. NGFW has all the features of the traditional firewalls, like packet filtering, network and port address translation (NAT), stateful inspection, and virtual private network (VPN). It also has more advanced features like intrusion prevention system (IPS), deep packet inspection, and user identification.

NGFW can identify and control traffic at the application layers by looking deep into the application layer, i.e., the packet’s payload. While traditionally, firewalls focus on the network protocols and the headers of the packets. Application intelligence is an essential part of NGFW. It means that NGFW is capable of identifying applications regardless of port and protocol, identifying a user and linking user identity with the packet, and identifying the true intention of the payload.

The concept of NGFW is to achieve everything a traditional firewall does with expanded capabilities that combine novel identification technologies, high-performance, and extra innovative features.

C. What NGFW Is Not

Just like mentioned in section II, many of the older versions of firewalls shared similar features with NGFW. Specifically,

UTM is just like an NGFW. UTM is a combination of all firewalls which provides multi-level protection against different threats. In the early days of NGFW, maybe some can say they are the same. Nevertheless, since then, NGFW has evolved into a more superior product. UTM targets a smaller environment, and it is not built for performance. In comparison, NGFW targets an enterprise-level environment and offers high performance.

Even though both Web Application Firewall (WAF) and NGFW inspect deeply into the application layer of the packet. They are not the same thing. WAF inspects only the application layer of the OSI model, while NGFW inspects all layers of the OSI model. WAF targets web applications, while NGFW targets the whole network. Hence NGFW offers more complete protection.

D. Key Features of NGFW

1) Application Awareness

Traditional firewalls use header information such as IP address, port number, transport protocol to identify the sender and the packet. However, there are some advanced evasive internet threats that apply evasive tactics such as port hopping, non-standard ports, SSL encryption, and tunneling protocol to bypass the traditional firewalls [14]. NGFW identifies the real application by looking past the packet header.

Traditional firewalls do not inspect the payload because the payload is protected by an application protocol, such as HTTP and FTP. NGFW decrypts the application protocol and re-encrypts it after the payload is analyzed, thus providing opportunity to look past application layer.

Most applications have known application signatures. They are unique transaction characteristics that would identify the application regardless of the protocol and the port. Any application with a suspicious signature is blocked by NGFW, and the ones allowed are examined further down. This way, evasive tactics such as port-hopping and non-standard ports can be detected and be handled.

Furthermore, the application that NGFW decrypted could be a front for what is hiding beneath it. In other words, the real application uses the first application protocol as a tunnel. So, how does NGFW determine if the application protocol belongs to the real application? This is where content inspection and heuristics analysis come in [15].

2) Content Inspection and heuristics analysis

The application layer of the OSI model is now the most accessible and exposed to the outside world; it is also the most difficult to defend since vulnerabilities identified here typically rely on complicated user input situations [16]. More than 70% of network attacks target the application layer [17]. That is why application layer protection is crucial and essential part of NGFW. Just like mentioned before, NGFW looks beyond the network layer and the transport layer. It dissects the payload of each packet and examines the application signature. It is sufficient to block most known attacks, but not all of them. Heuristic (or behavioral) analysis is the backup plan.

Heuristic analysis targets more complex applications that use unknown application algorithms or proprietary encryption, such as peer-to-peer file-sharing or VoIP applications. Heuristic analysis determines whether the application is harmful by observing the behavior of the program. Heuristic analysis typically uses three components to help with detecting harmful applications. The first component is the data collector. After the payload is decrypted from the application protocol, the data collector collects the dynamic and static information about the executable. Then comes the second component, the interpreter. It converts the information collected by the first component and transforms them into intermediate representations. The third component, matcher, compares the intermediate representations to a database that contains known harmful behavior signatures [18].

In addition to signature matching and heuristic analysis, URL and file filtering are also great tools for packet inspection. URL filtering restricts connection from certain network domains. File filtering allows NGFW to block files by their actual type (not based on just their extension) and the ability to control the transfer of sensitive data.

Combine signature matching, heuristic analysis, URL and file filtering. Network attacks which target the application layer are detectable and hence can be intercepted.

3) User Identification

Similar to the stateful firewall, NGFW keeps track of traffic flows and their corresponding senders and receivers. However, NGFW takes user identification to the next step. After NGFW obtains the basic user information from the header of the packet, such as destination IP address and port number, source IP address, and port number. NGFW communicates with the LDAP directories such as the Active Directory (AD), Microsoft's proprietary directory that links the user information with the user's IP address. By communicating with the AD, NGFW keeps its own user table up to date by frequently checking its own table against the AD. After the user-to-IP address relation is matched, NGFW requests more user information such as role and group assignment from the AD. By doing so, allows NGFW to gain clarity on who is responsible for a certain packet in the traffic flow and enable packet filtering by user identity [2].

User identification is especially important for network administrators. It facilitates troubleshooting and fast incidence response in case of a network emergency.

E. Benefits of NGFW

1) Performance

NGFW offers multiple performance increase on different aspect of its defensive layers. When it comes to traffic monitoring, NGFW uses stream-based packet scanning. An approach that drastically reduces latency and increases throughput by scanning the first packets as soon as they arrive, instead of waiting until the whole file is in memory, like traditional firewalls [19]. For example, in figure 8, it shows that the latency for both types of scanning methods is calculated by the difference between the finish time of the output process and the finish time of the payload receive process. However,

because stream-based scanning allows a quicker start time for the payload scanning process, thus latency is greatly reduced.

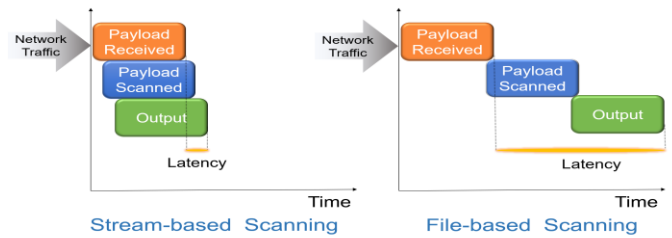


Fig. 8. Stream-based Scanning VS. File-based Scanning [20].

When it comes to threat signature comparing, NGFW uses uniform threat signature format. An approach that increases threat signature matching speed by avoiding the need to use separate scanning engines for each type of threat. Meaning viruses, spyware, worms, Trojans, adware, and other threats can all be detected in a single pass [20].

When comparing NGFW to other firewalls, UTM is the one that is mentioned the most because they do share many similarities. However, the most significant difference between UTM and NGFW is their performance. When a packet reaches UTM, it is passed around by different smaller firewalls, meaning the packet is scanned by many independent components inside UTM. Many of these scans are repetitive and inefficient. Thus causing a massive performance decrease. In figure 9, it shows that the traditional multi-pass architectures for an UTM. Specifically, each defense component runs after the previous components are finished with the packet. Each column in figure 9 represents different type of defense module. For example, column one represents packet filtering module, in which a packet has to be scanned by three components. After the packet is scanned by the first module, it is passed to the second module, which is represented by column two. Column two in this case represents the web defense module, which consists of four components. Column three represents the intrusion prevention system (IPS) module, and column four represents the anti-virus (AV) module. A packet has to go through all these components for scanning before the firewall decide if the packet is safe or not. In the real world, there could be many more modules and that is why traditional multi-pass architectures exist in an UTM is considered inefficient and is being replaced by NGFW.

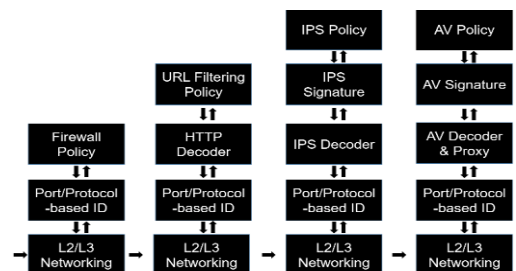


Fig. 9. Traditional multi-pass architectures [2].

NGFW is fundamentally different from this multi-pass architecture. NGFW was built on a single-pass architecture. Every component talks to each other and the scans are done in

parallel processing. Figure 9 shows the single-pass architecture used designed by Palo Alto Networks, it is used in their NGFWs. The architecture itself allows multiple copies of packets to be distributed to different scanning modules and they all process the packet in a parallel fashion. By doing so, eliminates repetitive packet handling, minimizes wait time, and maximizes throughput and performance.

Each row in figure 10 represent a defense module. Unlike traditional multi-pass architecture shown in figure 9, the component inside a defense module runs in parallel fashion. For example, the application identification module is responsible for identifying the application via different methods. When a packet reaches this module, copies of the packet is made. Each component runs their algorithm simultaneously. Therefore, there is no wait time between components which greatly reduce the processing overhead.

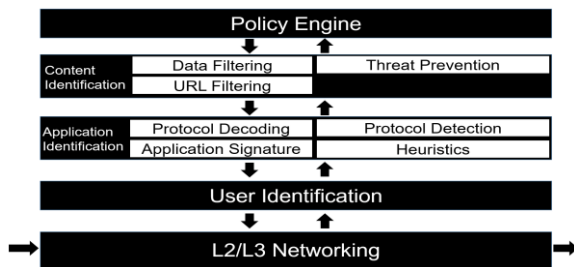


Fig. 10. Single-pass architecture by parallel processing.

2) Security

The concept of NGFW is that it is everything traditional firewalls can do, plus wider and deeper protections. Thus, NGFW operates at layer 3, layer 5 and layer 7 of the OSI model. At layer 3 which is the network layer, NGFW uses traditional firewall technique such as packet filtering by IP address or by port number to intercept malicious packets. At layer 5 which is the session layer, NGFW serves as gateway between two devices, inspecting each data segment before it reaches the receiving device. At layer 7 which is the application layer, NGFW process a proven threat detection and prevention system which include the following mechanisms:

Protocol decryption and re-encryption allows NGFW to inspect packet at layer 7 of the OSI model. Hence making application layer attacks detectable.

- Anomaly-based detection allows NGFW to detect non-RFC connections such as overlong URI and overlong FTP login.
- Statistical anomaly detection allows NGFW to prevent rate-based DDoS flooding attacks.
- Signature-based analysis allows NGFW to prevent known attacks with a minimal error rate.
- Huestis-based analysis allows NGFW to prevent attacks bypassing signature-based analysis.
- Sandbox Integration, which allows NGFW created an isolated and safe environment to test and monitor any suspicious executable without endangering an organization's network.

- Cyber threat intelligence (CTI), the action of assessing, choosing, and recording threat information data based on the needs of relevant companies in order to generate unified threat intelligence. CTI allows NGFW to update threat signatures more frequently and more accurately [21].

In addition to the IPS/IDS system, NGFW also provides important defense mechanisms such as anti-spam and reputation and identity services that can help better protect the network.

3) Simplicity

It can be a headache when it comes to managing a traditional firewall due to its complexity and lack of a graphical user interface (GUI). NGFW offers enhanced visibility and control by providing a GUI, which allows enterprises to focus on more relevant elements rather than try to remember the correct commands.

NGFW also reduces the complexity of the process of managing a firewall. So, the enterprises can focus on a simpler concept such as usernames or application names rather than IP address or port numbers.

F. Vendors

There are many different versions of NGFW, each supported and developed by a unique vendor. Cisco is one of the leading companies in developing a reliable NGFW. The Cisco Firepower™ Next-Generation Firewall comes with all the essential components for NGFW such as IPS, DPI, threat intelligence sources. In addition, due to the fact that it was developed by Cisco, one of the largest security conglomerate corporations in America. Cisco Firepower also provides excellent integration services with other tools from Cisco. According to Gartner Peer Insights, Cisco Secure Firewall has an average score of 4.4/5 with 892 reviews. The highest evaluations and ratings for Cisco were for the quality of technical assistance and the timeliness of vendor answers [22].

Palo Alto Networks is another vendor of NGFW. Unlike Cisco, Palo Alto Network is a rather new company and was founded solely for developing NGFW. One of its founders, Nir Zuk, was one of the few people who helped develop the earliest stateful inspection firewall and IPS. Due to their concentration in developing NGFW, Palo Alto Network is the first company that provides NGFW powered by machine learning. Which aims to protect the industrial network in all aspects, including but not limited to phishing attacks, zero-day attacks, IoT security, and policy management [23].

FortiGate, an NGFW supported by Fortinet, is a secure and fast responsive device that serves a range of clients from the home office to distributed enterprise organizations and data centers. Inside the device, the Secure Processing Unit (SPU) works as a CPU co-processor by taking most resource-intensive security functions such as application identification, IPS, DPI, so the CPU can perform other important tasks. FortiGate focuses on operating system protection, especially with FortiOS, an operating system that is security-hardened, purpose-built in which focuses on system and network security protection. FortiGate is fully integrated with FortiOS. By using

both FortiGate and FortiOS, the network is protected from both software and hardware levels.

There are many more NGFW vendors out there, such as Juniper Networks, Huawei, Sophos, and Forcepoint. All of them provide a robust and unique version of NGFW. But the three mentioned above are the ones I find most interesting.

IV. NGFW APPLICATIONS

A. Machine-Learning and NGFW

As we mentioned in section III, the intrusion prevention system in NGFW depends on two types of detection methods for network threats, signature-based and heuristics-based. Signature-based analysis depends on finding the features of the threats. In comparison, heuristics-based analysis depends on finding the behavioral pattern of the threats. Finding out the features and the patterns comes with a high cost when is done by human. Recent developments in machine learning have impacted the ways we identify network threats. Namely, we are using data mining and machine learning techniques to learn the behaviors and the features of a malicious executable file. The very first attempt was made by Multi Naïve Bayes, which was to classify malware and benign files [24].

Besides IPS, machine learning could also be useful in a different component of NGFW. Cisco have recently developed a neural network that detects Advanced Evasion Techniques (AETs). AETs are done by threat actors, which try to cover up their malware or viruses. AETs are methods that use different network protocols and associated techniques (e.g., DNS tunneling) across OSI layers 3–7 to evade detection by the standard perimeter security stack (IPS, NGFW, etc.). The neural network was trained by feeding training data and compromise indications from Intrusion Prevention Systems (IPS) and Next-Generation Firewalls into deep learning unsupervised machine learning algorithms [25]. With the help of the trained neural network, Cisco NGFW is able to detect harmful packets that contains virus or malware from what seems like a legitimate traffic.

This is an active area of research and there are many more ongoing researches regarding network security and machine learning such as a network anomaly detection by random forest approach [26], malicious traffic packet classification by machine learning [27], and so on.

B. NGFW In IoT

Information technology (IT) is rapidly growing due to the growth of the Internet of Things (IoT). Unlike a decade ago, only certain types of electronics were connected to the internet. Nowadays, everything is a smart device and connected to the internet. Products like TVs, cameras, refrigerators, air conditioners, thermostats, microwaves, and so on. Some of them are connected through a wire, such as an ethernet, and some of them are connected wirelessly by Wi-Fi. The more devices are connected to the router, the more exposure in the network, hence more vulnerabilities and risks to the network. The important question is how to protect our exposed network effectively.

As mentioned in the previous chapters. Traditional firewalls do not care about application-layer protection. Application

layer attacks such as worm attacks and side-channel attacks which target IoT devices are quite common these days. They only need one node as an access point and can cause serious damage to the entire network. Malicious node injection attacks become more difficult to detect as the number of malicious nodes increases [28].

According to research done by Soewito, Benfano, and Charlie [29]. They created two separate IoT networks, one is protected by a traditional firewall named IBM ISS Proventia, and the other is protected by an NGFW, called Checkpoint. Attacks such as phishing attacks, DDoS, and SQL injections are launched on both IoT networks. A DDoS attack by TCP-SYN flooding against the NGFW protected network was dropped by the NGFW, thus was harmless against the network. On the other hand, the traditional firewall protected network allowed this packet to pass and caused heavy CPU usage on the server. A second DDoS attack by UDP flooding was intercepted by both firewalls and did not do any damage. The third attack, SQL Injection, was rejected by NGFW and allowed by the traditional firewall. The fourth attack, phishing by email with a virus, was rejected by NGFW and allowed by the traditional firewall. The result of this case study concludes that traditional firewall protected IoT network is only safe from layer 3 to 4 attacks, while NGFW protected IoT network is safe from layer 3 to 7 attacks. In other words, traditional firewalls are no longer capable of protecting an exposed IoT network, only NGFW could offer all around protections in today's IoT network.

V. CONCLUSION

Next-generation firewalls are the next step up from traditional firewalls. Traditional firewalls generally perform packet forwarding and blocking, which determine only by header information. Due to a lack of packet inspection, they are powerless against application-layer attacks. Other newer firewalls, such as application-gateway firewalls and web-application firewalls, provide packet inspection. However, they lack basic protection provided by traditional firewalls. Combining traditional firewalls and newer firewalls gives us UTM, but due to the multi-layer protection, the packet must travel through many layers of scanning to reach the final destination, which causes a drastic performance decrease. NGFW provides all layers of security protection without the loss of performance. Although there are many different companies offering NGFWs, each with its own set of advantages and disadvantages, the concept of combining the functionality of a traditional firewall with IDS and the ability to manage traffic at the application layer without decreasing network performance will define the future standard for enterprise and industrial network protection.

REFERENCES

- [1] Ingham, Kenneth, and Stephanie Forrest, "A history and survey of network firewalls", University of New Mexico, Technical Report, 2002, available on www.unm.edu, accessed on Dec 28, 2021
- [2] Lawrence, C., "Next Generation Firewalls for Dummies", UNLV Press, 2011
- [3] David Smith, "2020 Data Breaches - the Most Significant Breaches of the Year: IdentityForce. We Aren't Just Protecting You From Identity Theft. We Protect Who You Are"., Network Magazine, April 15, 2021.

- Available on <https://www.identityforce.com/blog/2020-data-breaches>, accessed on Dec, 28, 2021.
- [4] Schultz, E. Eugene. "83-10-41 Types of Firewalls" Internet: <http://www.ittoday.info/AIMS/DSM/83-10-41>. accessed on Dec, 28, 2021.
 - [5] Wool, Avishai. "Packet filtering and stateful firewalls" Handbook of Information Security 3 (2006): 526-536
 - [6] "Application Gateway" Network Encyclopedia, May 3, 2020. <https://networkencyclopedia.com/application-gateway/>. accessed on Dec, 28, 2021.
 - [7] Gupta, Namit, Abakash Saikia, and D. Sanghi. "Web application firewall" Indian Institute of Technology, Kanpur 61 (2007): 62.
 - [8] "WAF vs. NGFW: Which Technology Do You Need" F5 Networks. Accessed September 30, 2021. <https://www.f5.com/c/landing/waf-vs-ngfw-which-technology-do-you-need>, accessed on Dec, 28, 2021.
 - [9] "Web Application Firewall Market Share, Size and Industry Growth Analysis 2021 - 2026." IndustryARC. <https://www.industryarc.com/Research/Web-Application-Firewall-Market-Research-502436>, accessed November 9, 2021.
 - [10] Cruz, Luis. "What Is Unified Threat Management (UTM)?" Study.com. <https://study.com/academy/lesson/what-is-unified-threat-management-utm.html>, Accessed November 9, 2021
 - [11] Agham, Vinit. "Unified threat management" International Research Journal of Engineering and Technology 3, no. 4 (2016): 32-36.
 - [12] Todd McGuiness. "Defense in Dept". sans.org. Archived from the original on 22 Dec 2017. Retrieved 22 December 2017.
 - [13] "The Evolution of Internet Security." Preferred IT Group, LLC.. <https://www.preferreditgroup.com/2018/05/11/the-evolution-of-internet-security/>, accessed November 10, 2021
 - [14] Matt Keil September 8, Matt Keil, and Matt Keil. "Balancing the Risks and Benefits of Evasive Applications." Palo Alto Networks Blog, September 8, 2009. <https://www.paloaltonetworks.com/blog/2009/09/controlling-evasive-applications/>, accessed on Dec, 28, 2021.
 - [15] "App-ID Technology Brief - Palo Alto Networks." Accessed November 13, 2021. <https://media.paloaltonetworks.com/documents/techbrief-app-id.pdf>, accessed on Dec, 28, 2021.
 - [16] Amos, Jesse Daniel. "7 Layers of Cybersecurity Threats in the ISO-OSI Model." Computer Learning Courses - Online & In Person Training. Accessed November 13, 2021. <https://training.nhlearninggroup.com/blog/7-layers-of-cybersecurity-threats-in-the-iso-osi-model>, accessed on Dec, 28, 2021.
 - [17] Koch, Robert. "Towards next-generation intrusion detection." In 2011 3rd International Conference on Cyber Conflict, pp. 1-18. IEEE, 2011.
 - [18] Bazrafshan, Zahra, Hashem Hashemi, Seyed Mehdi Hazrati Fard, and Ali Hamzeh. "A survey on heuristic malware detection techniques." In The 5th Conference on Information and Knowledge Technology, pp. 113-120. IEEE, 2013, pp. 2
 - [19] "Network-Based Malware Protection." Palo Alto Networks - network-based malware protection - NextGig Systems, Inc.. Palo Alto Networks Reseller. https://www.nextgigsystems.com/nextgen_firewall/palo_alto_antivirus.html, accessed November 13, 2021.
 - [20] "Integrated Threat Prevention - Paloguard." https://www.paloguard.com/datasheets/Threat_Prevention_ds-new.pdf, accessed November 13, 2021.
 - [21] Guo, Liang, Senhao Wen, Dewei Wang, Shanbiao Wang, Qianxun Wang, and Hualin Liu. "Overview of Cyber Threat Intelligence Description." In International Conference on Applications and Techniques in Cyber Security and Intelligence, pp. 343-350. Springer, Cham, 2021.
 - [22] Gartner, Inc. "Cisco Secure Firewall Reviews, Ratings, and Features - Gartner 2021." Gartner. accessed November 30, 2021.
 - [23] Palo Alto Networks Santa Clara, CA Jun 17. "Palo Alto Networks Launches World's First ML-Powered NGFW." Palo Alto Networks. <https://www.paloaltonetworks.com/company/press/2020/palo-alto-networks-launches-worlds-first-ml-powered-ngfw>, accessed November 30, 2021.
 - [24] M. Schultz, E. Eskin, E. Zadok, and S. Stolfo, "Data mining methods for detection of new malicious executables." In IEEE Symposium on Security and Privacy, pages 38-49. IEEE COMPUTER SOCIETY, 2001.
 - [25] Badhwar, Raj. "Polymorphic and Metamorphic Malware." In The CISO's Next Frontier, pp. 279-285. Springer, Cham, 2021.
 - [26] Hajialian, Hesamaldin, and Cristian Toma. "Network anomaly detection by means of machine learning: random forest approach with apache spark." Informatica Economica 22, no. 4, pp. 89-98, 2018
 - [27] Suga, Toki, Kazuya Okada, and Hiroshi Esaki. "Toward real-time packet classification for preventing malicious traffic by machine learning." In 2019 22nd Conference on Innovation in Clouds, Internet and Networks and Workshops (ICIN), pp. 106-111. IEEE, 2019.
 - [28] Deogirikar, Jyoti, and Amarsinh Vidhate. "Security attacks in IoT: A survey." In 2017 International Conference on I-SMAC (IoT in Social, Mobile, Analytics and Cloud)(I-SMAC), pp. 32-37. IEEE, 2017.
 - [29] Soewito, Benfano, and Charlie Erwin Andhika. "Next generation firewall for improving security in company and iot network." In 2019 International Seminar on Intelligent Technology and Its Applications (ISITIA), pp. 205-209. IEEE, 2019.